

Class 5: Data Viz with ggplot2

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Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the ‘plot()’ function and add on packages like **ggplot2**.

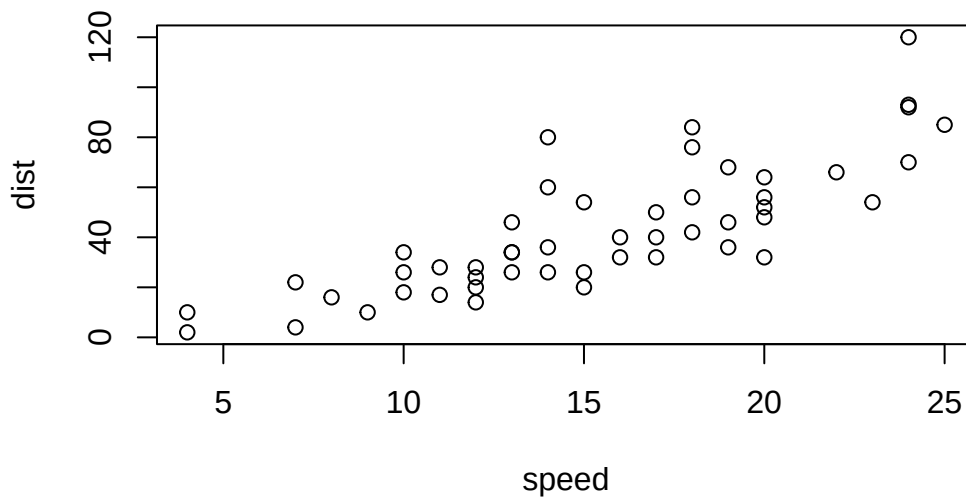
Let’s compare how we make a simple figure with these two systems:

We can use the in-built **cars** dataset:

```
head(cars)
```

```
      speed dist
1         4    2
2         4   10
3         7    4
4         7   22
5         8   16
6         9   10
```

```
plot(cars)
```



Before I can use `ggplot2` I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

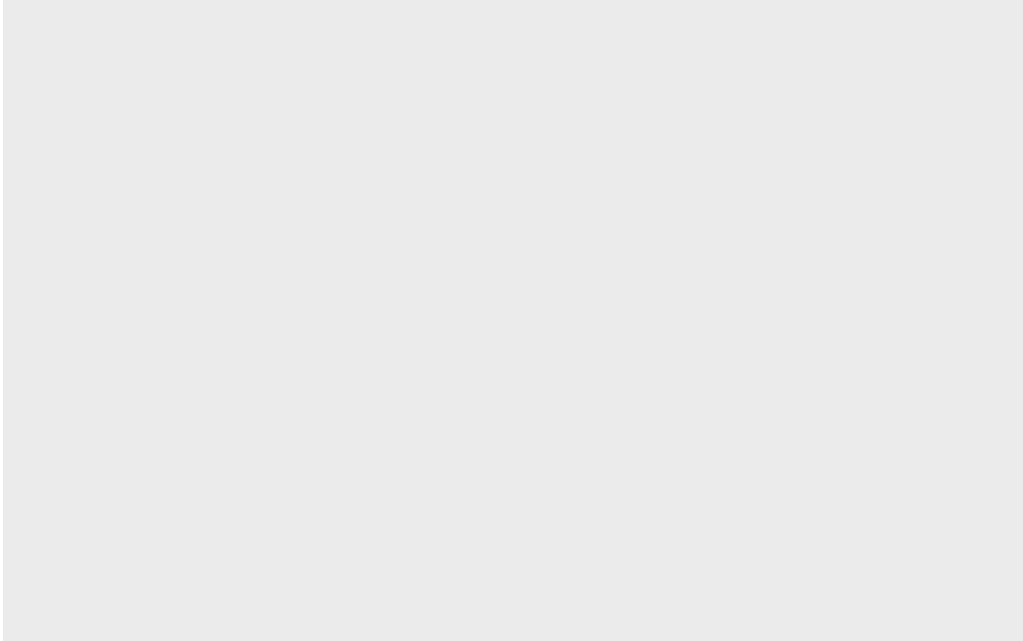
N.B. we never run `install.packages()` in our quarto doc (we run it once only in our R console) as it would re-install the package every time we render our quarto report.

Once installed we need to load up the package into our R brain:

```
library(ggplot2)
```

The main function in the **ggplot2** package is called `ggplot()`

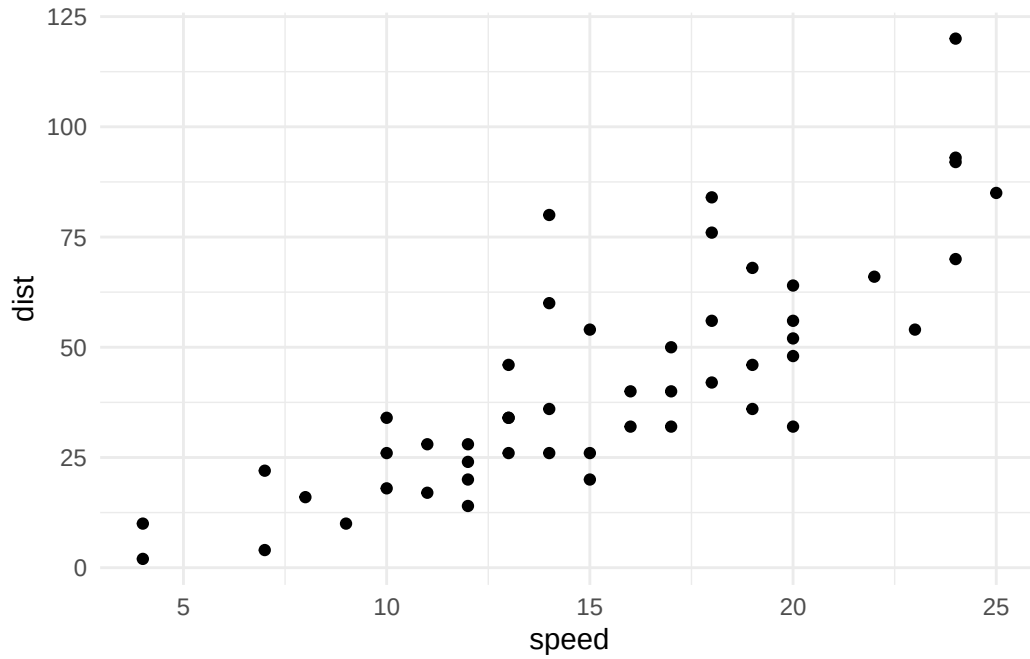
```
ggplot(cars)
```



Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot),
- the **geom** layer (how you want the plot drawn, e.g. points, lines, histogram, etc)

```
ggplot(cars) + aes(x= speed, y=dist) +geom_point()+theme_minimal()
```

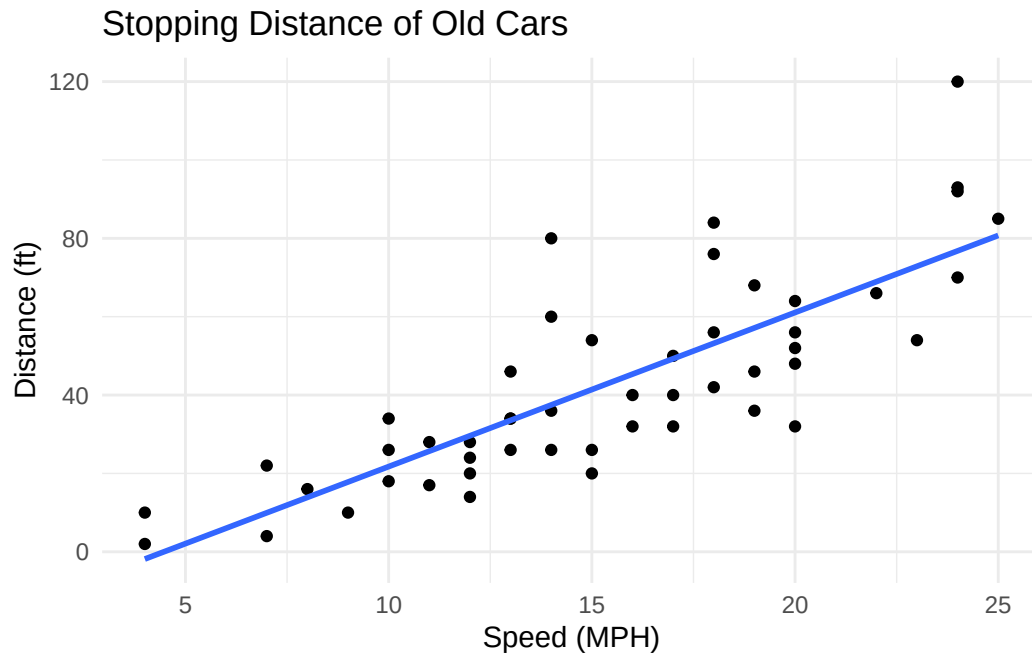


```
##add some custom features
```

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) + aes(x= speed, y=dist) +geom_point()+theme_minimal()+ geom_smooth(method = "lm"
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Q: can you make the `geom_smooth()` function produce a linear straight line fit to the data and turn off the “gray” error region. A: use `geom_smooth(method = “lm”, se = FALSE)`

Gene Expression figure

Import the data to plot

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

```
sum(genes$State == "up")
```

```
[1] 127
```

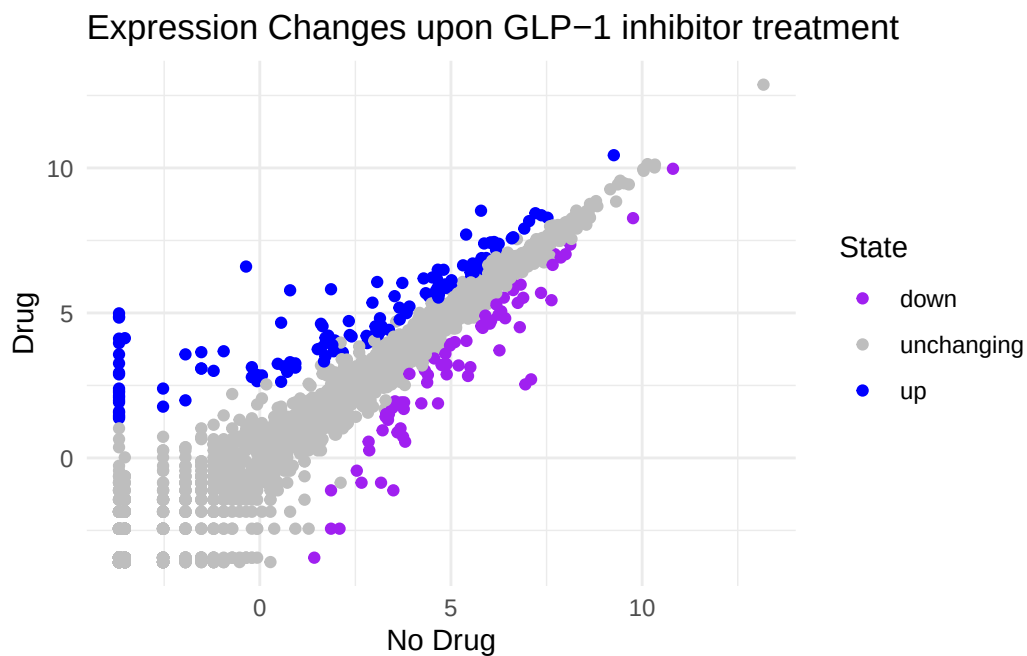
A useful new function in this context is the `table()` function:

```
table(genes$State)
```

```
down  unchanging    up
   72      4997    127
```

My first plot attempt

```
ggplot(genes)+aes(Condition1,Condition2,col=State)+geom_point() + scale_colour_manual( values
```



Going Further

Here we read the famous gapminder dataset:

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Q. how many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different “country” entries are in this dataset?

```
length(unique(gapminder$country))
```

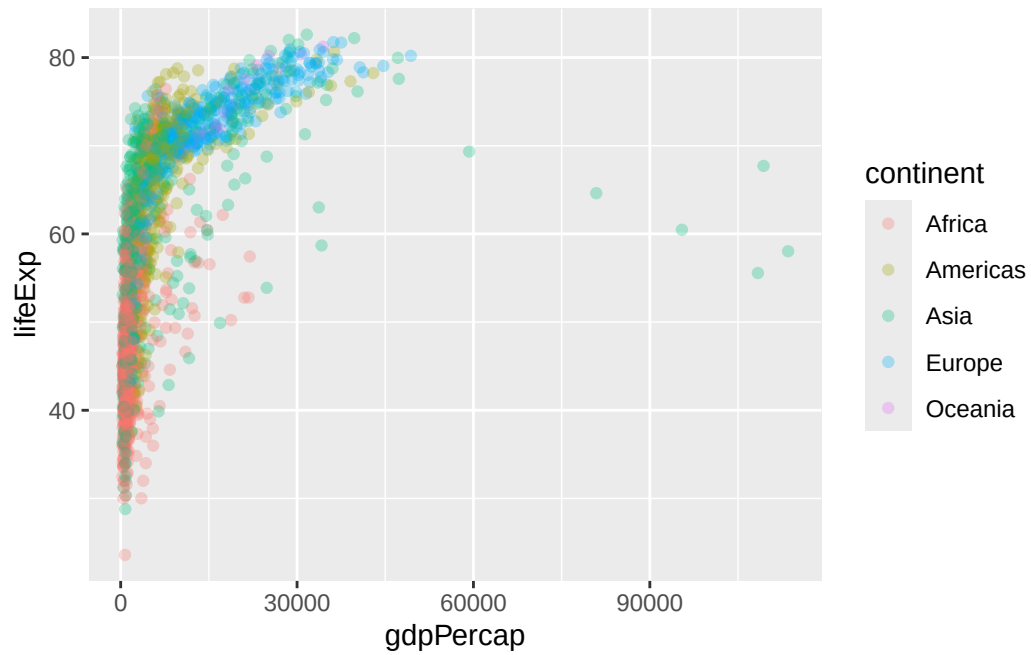
```
[1] 142
```

Let’s make our first plot of the entire dataset:

Plot of “gdpPercap” vs “lifeExp” colored by “continent”

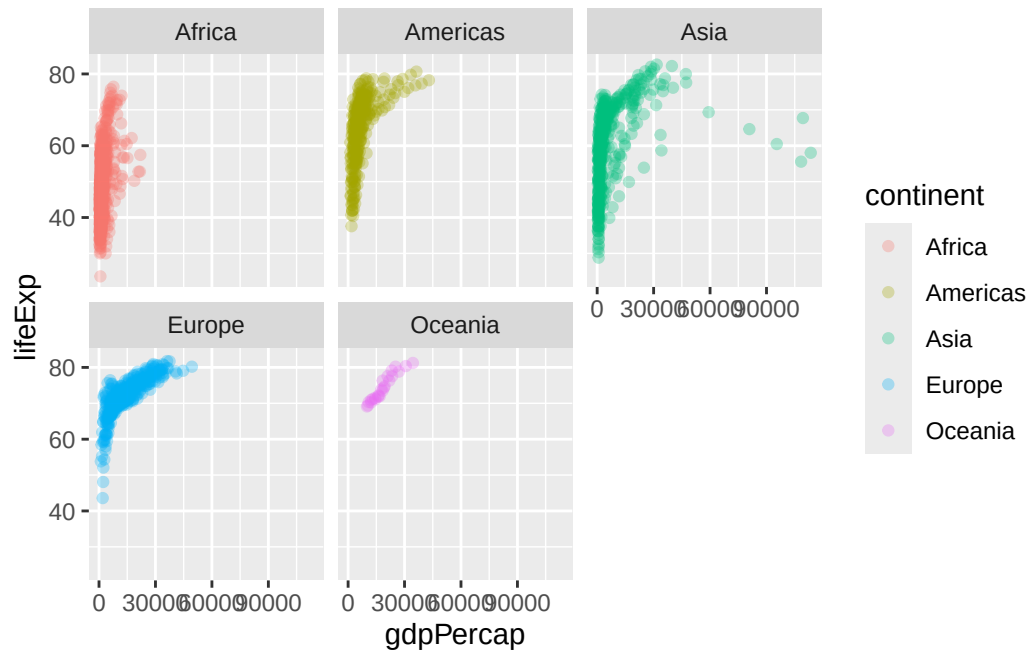
```
p<-ggplot(gapminder)+aes(gdpPercap, lifeExp, col=continent)+geom_point(alpha=0.3)
```

```
p
```



I can add more layers using `p`

```
p + facet_wrap(~continent)
```



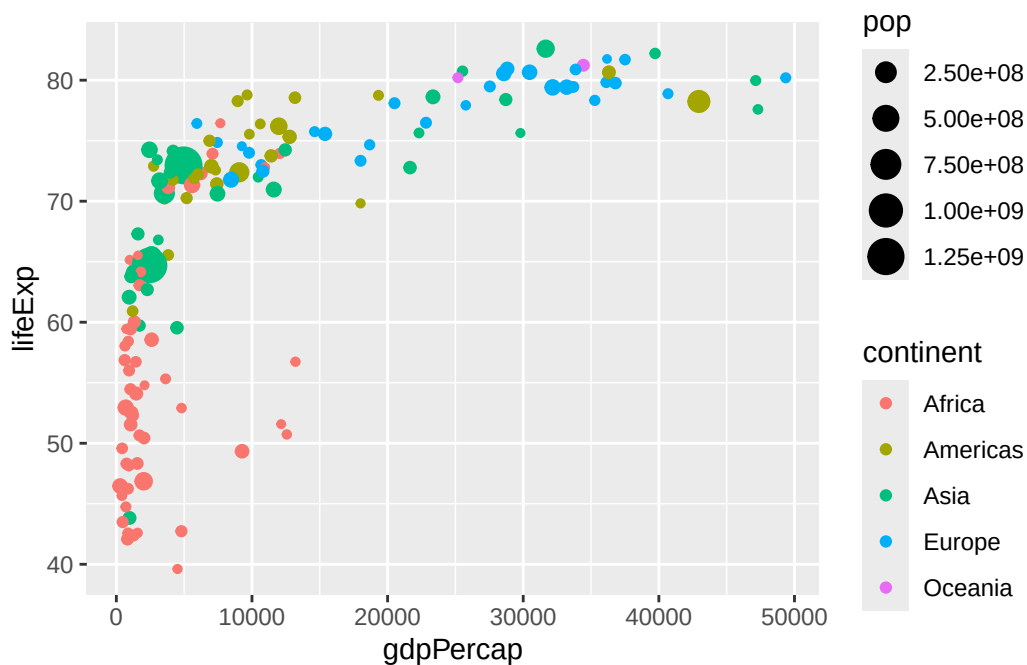
Make a plot for years 1977 and 2007 only (not all the yaers in the dataset.)

Q. First use the **dplyr** package and the **filter()** function from that package to extrat the year 2007

```
library(dplyr)
```

```
g07<-filter(gapminder,year == 2007)
g77<-filter(gapminder,year == 1977)
g<-filter(gapminder,year == 2007 | year == 1977)
```

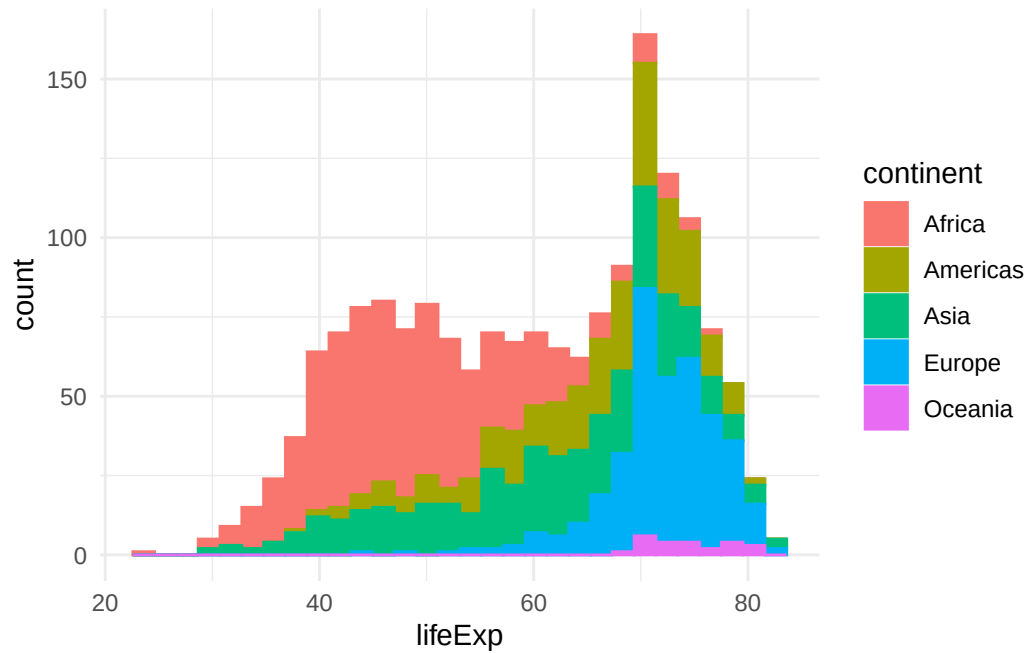
```
ggplot(g07)+aes(gdpPercap,lifeExp,col=continent,size=pop)+geom_point()
```



Q. Make a histogram of lifeExp colored by continent (try using **fill=continent** or **col=continent**) Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder)+aes(lifeExp,col=continent,fill = continent)+geom_histogram()+theme_minimal
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
ggplot(gapminder)+aes(lifeExp,col="black",fill =continent  
)+geom_histogram()+facet_wrap(~continent)+theme_minimal()
```

``stat_bin()` using `bins = 30`. Pick better value `binwidth`.`

